S3 Text. Protocol deviations and independent verification

Part A. Protocol deviations (verbatim from docs/DISCLOSURES.md)

Collected at M3 (T+14h15) by an independent retrospective agent; maintained by the PI.

1. **The TreeFam-A dev split was created after QfO peek 1** (peek 1 cafe3f5 → split f65d5b7). Peek 1 had already shown aggregate TreeFam-A and SwissTree recall, and that motivated the recall work. So dev/test is not blind in the DESIGN §4 sense. CORRECTION (verification pass 2): TreeFam-A **test-family** scores WERE computed at the logged peek 2 (results/qfo_peek2/treefamA_dev_test.json; accurate .580/.913, fast .605/.914), on PI authorisation, before the freeze tag. They did not drive any tuning (the presets were fixed before peek 2 and only the freeze followed), but dev/test is not blind.
2. **Heavy dev reuse:** OrthoBench dev has 243 runs on 34 RefOGs, and synteny parameters, local trees, FastTree, the presets and deep absorption were all selected on it. Dev numbers (79.7 vs OrthoFinder 70.5) are optimistic; the comparator was not tuned. Claims rely on OrthoBench **test** only.
3. **Deep-node threshold** (median seed $-\ln S \geq 1.5$) was defined and tuned on QfO TreeFam-A dev. No OrthoBench node qualifies, so it is QfO-specific and untested on OrthoBench.
4. **The S0 completeness target (≥95%) was missed:** dev 0.794, test 0.722, against a sequence-similarity ceiling ≈0.82. Recorded as a deviation; test completeness was logged.
5. **Non-inferiority on recall failed at peek 1** (TreeFam-A TPR .55 vs OrthoFinder .74); the final result must be reported against the pre-registered rule as it stands.
6. **Comparator rows** are published OpenEBench values, produced under other conditions. FastOMA benefits from the OMA database. Our first local FastOMA OrthoBench run (dev 29.6) was at first thought misconfigured. It was later traced to a FastOMA 0.5.1 gene-dropping bug with correct inputs (see #19).
7. **Design review round 3 was not run** (deferred, then superseded by data-driven reviews). The freeze moved from T+36h (DESIGN §7) to T+20h, to leave time for evaluation and review.
8. **Post-hoc redesigns.** Constrained synteny replaced the additive bonus after it failed. The micro → macro pair-F1 switch was made on dev. Shuffled-control spans are ±1–3 OG-F, so OG-level synteny gains of +4–7 are moderate evidence. VGNC PPV .982 suggests vertebrate over-merging.
9. **No claim was independently verified before M3.** Independent verification of CLAIMS.md is scheduled after the freeze. G-STD2/STD tests are run once, on the frozen config.
10. **Adoption rules were first checked only on the fast preset.** so_struct passed on fast and failed on accurate; the presets now differ in options (each adopted on its own base).
11. **The merge output depends slightly on Newick child order** (same topology, different child order → 18 OrthoBench OGs differ). Determinism holds for a fixed input file. Found in the final evaluation, after the freeze; not fixed.
12. **The synteny OG-level gain on OrthoBench dev (+4.4 OG-F) did not replicate on OrthoBench test** (−0.1 when removed). Report the synteny effect as dev-only at OG level; pair-level and QfO effects separately.
13. **Early exposure of test RefOGs** (found by independent verification): before the homology-component split, three S0 configs (commit 6711002) were scored per RefOG on the old random split's dev half; 16 of those RefOGs are in the current test set. Impact is likely small (only early S0 choices; the new split was rule-based with a fixed seed), but test is not perfectly blind.
14. **Runtime comparison is loose:** Ancestra ran at 10–12 threads, competitors at 8, on a shared node (load 20–65); DESIGN asked for 24 threads on an idle node. CPU-h is the fairer comparison. Gene-order table building (one-off parsing of GFF) is not included in Ancestra's 3.9 CPU-h.
15. **Exploratory status:** the 11 OrthoBench-test ablations and all QfO ablations have no multiple-testing correction; they are exploratory. "Precision comparable" at QfO peek 1 should read "inconclusive (no CI)".

The S4 verdict "keep NJ" was superseded by the later FastTree merge arm (adopted on dev by its own rule).

16. **Commit-ID bookkeeping:** freeze.json was written one commit before the freeze tag; the code (ancestra/*.py) is byte-identical across 6c593a8, 71117bf (tag freeze-v1) and f7bcddc (cited in test_log).
17. **The QfO ablation CIs reported by the official test (±0.006) ignore within-family clustering of pairs.** Family-level bootstrap (1,108 families) gives: no gene order −0.012 [−0.028, +0.001]; ASTRAL-Pro +0.028 [−0.009, +0.077]; FastME +0.022 [−0.001, +0.045]. So no gene-order effect on QfO recall is supported.
18. **Wording corrections:** "STD comparable" had no CI; on STD-Euk, Ancestra is worse than OMA, Hieranoid and InParanoid. VGNC PPV (.982) is not lower than OMA's (.979–.980). The GO/EC "±" in qfo_final/summary.md is 1.96×SEM, not SEM. test_log commit IDs differ from the run manifests (the code is identical).
19. **OMA comparators were added after the freeze at user request** (post-freeze, compared with frozen outputs only). The OMA-browser-HOG OrthoBench conversion was first scored with a wrong labelling rule (test 45.5) and then corrected (test 60.4). Both are recorded in bench_repo/results/orthobench_competitors/oma_browser/. OMA contains all OrthoBench species (leakage favours OMA). FastOMA 0.5.1 drops ~5.5k genes when splitting groups (bug). We patched it and re-ran (gene coverage 79.0%→80.6%): test F 33.1 (v1 33.5), so the bug does not explain FastOMA's low score. The corrected OMA-browser labelling rule was justified by inspecting RefOG005, but it was chosen after seeing a score and moved the result in OMA's favour, so it is conservative for Ancestra.

Part B. Independent verification report (verbatim from docs/VERIFICATION.

Independent verification (post-freeze, freeze-v1)

Verifier: independent agent (produced none of the results), 2026-10-27, T+17h29 to T+17h45 (scripts/clock.sh). Scope: OrthoBench final test, reproducibility, toy end-to-end, 20-HOG spot check, and one recomputed number for each of S0, S1, S4, S5 and S8. All verifier code and outputs are in `results/verification/`. Nothing in `ancestra/`, `CLAIMS.md` or other agents' results was modified. Threads ≤ 6 throughout.

1. Claim table

claim (CLAIMS id)	reported	recomputed	match?	how / notes
OB **test** F, Ancestra accurate (final)	76.2 (P 69.5, R 84.2, 10/36 exact)	76.19 (P 69.54, R 84.23, 10 exact)	yes	`ob_rescore.py`: official `benchmark.py` readers + `calculate_benchmarks_pairwise` called on the 36 test RefOG lists; does not import `bench_repo/scripts/orthobench` Input: raw `work/ob_final/main_accurate_g`
OB dev / all-70 F, accurate	79.7 / 77.9	79.69 / 77.90	yes	Official CLI on the raw file, all 70: 77.9 F, 72.5 P, 84.1 R, 21 exact (`official_cli_all70.txt`)
OB test F, fast	74.8 (P 68.7, R 82.1, 9/36)	74.79 (68.68 / 82.09, 9)	yes	dev 75.71, all 75.24 (CLI 75.2)

claim (CLAIMS id)	reported	recomputed	match?	how / notes
OB test F, OrthoFinder 3.2	69.7	69.70	yes	from raw `Orthogroups.og`; CLI on the raw OrthoFinder `Orthogroups.tsv`: all-70 70.1 / 19 exact = bench_repo
OB test F, SonicParanoid2 / ProteinOrtho	70.4 / 47.5	70.45 / 47.49	yes	raw `.og` files
ΔF test accurate − OrthoFinder	+6.5 [−6.5, +18.1], SE 6.4, NI not shown	+6.48 [−6.49, +18.05], SE 6.38	yes	own paired bootstrap (B=2000, seed 11). Robustness: seeds 23/47/1234 with B=10,000 give [−5.8…−6.0, +18.6]; leave-one-RefOG-out Δ ranges +3.2…+9.5; per-RefOG 15 wins / 12 ties / 9 losses. **Verdict "not shown non-inferior" is robust**
ΔF test accurate − SonicParanoid2	+5.7 [−4.3, +14.9]	+5.74 [−4.29, +14.86]	yes	other seeds: lower bound −3.8 to −4.0 (still below −3)
ΔF test accurate − ProteinOrtho	+28.7 [+13.1, +41.9]	+28.70 [+13.12, +41.94]	yes	
ΔF test fast − OF / SP2 / PO	+5.1 [−7.5,+16.6] / +4.3 [−6.0,+14.1] / +27.3 [+12.5,+40.1]	+5.09 [−7.49,+16.61] / +4.34 [−6.01,+14.07] / +27.30 [+12.53,+40.12]	yes	
Test-split IDs	36 test RefOGs	per_refog.tsv test labels == `splits/orthobench_split.json` test list; dev ∩ test = ∅	yes	`sha256sum -c splits/SHA256SUMS`: all 3 OK

claim (CLAIMS id)	reported	recomputed	match?	how / notes
No OB test score before freeze except S0 completeness	claimed	none found	yes (one caveat, §2.2)	`git grep` of the freeze-v1 tree and `git log -p` of the history for all 36 test RefOG IDs; scan of untracked pre-freeze files in results/ work/ logs/. Only `results/S0_families/2sA_dc30_` (logged). `orthobench_dev_eval.py` refuses non-dev subsets. Final per-config scores were written 08:05, after the 06:40 test_log entries
freeze.json hashes == working tree	24 files	24/24 identical; `git diff freeze-v1 HEAD -- ancestra/` empty	yes	see §2.3 for a commit-id inconsistency
pytest (not slow)	156 tests (tree)	**156 passed, 1 deselected**, 86 s	yes	`pytest_not_slow.log`
Toy e2e (C-CORE-1, pair F1 0.94–0.98)	0.94–0.98	seed 23, 8 sp × 300 genes: accurate 0.961 (P 0.993, R 0.931), fast 0.963; with gene order 0.961 / 0.963	yes	`toy_e2e.json`. hogs.orthoxml **byte-identical** across 2 runs per preset (with and without gene order) and across 1/4/6 threads
OB output validity (C-DOC-2 analogue)	PASS	`verify_orthoxml.py` on main_accurate_go: all 8 checks PASS (0 TaxRange≠LCA, 0 nesting violations, QfO flattening identical on 308,802 sampled pairs, pyham full load 15,246 top-level HOGs)	yes	own check: top-level groups (13,669 orthologGroups + 1,632 paralogGroups) re-derived from the OrthoXML == the 15,301 multi-gene OGs in `orthobench_ogs.txt` exactly; 20-HOG subset loads in pyham **without** the compat rewrite (20/20)
C-S0-2 S0 dev completeness	0.794 (27/34)	0.7941 (27/34)	yes	`s0_completeness.py`, own code on `work/S0/2sA_dc30_cc_sh30_RA/f` (sha256 = manifest). Keep-uncertain: 0.765

claim (CLAIMS id)	reported	recomputed	match?	how / notes
C-S0-4 S0 test completeness	0.722	0.7222 (26/36)	yes	
C-S5-2 synteny AUC human/mouse w=20	0.92	0.9223	yes	`s5_auc.py`, Mann–Whitney AUC from `work/S5/pair_scores_*.parquet` also human/chicken 0.928, zebrafish/Tetraodon 0.817, human/zebrafish 0.696, human/fly 0.488, shuffled (seed 11) 0.48–0.51; all RefOGs in dev. Labels themselves not re-derived
C-S8-2 hidden-gene recovery	92.2% [89.7, 94.1]	553/600 = 92.17%, Wilson [89.7, 94.1]; 0/512 random windows intact	yes	`s8_controls.json`; per species 95/96/98/90/89/85 as doc
S4 label accuracy NJ vs FastTree (c3di, matched)	0.928 vs 0.955, Δ +0.027 [+0.004, +0.060]	0.9277 vs 0.9551, Δ +0.0274 [+0.0038, +0.0591]	yes	`s4_acc.py`, own pooling + bootstrap over 32 sets from `per_set.tsv` (full scope: +0.034 [+0.019, +0.062]); all sets dev. Pair labels were not re-derived from the Newick files
C-S1-1 NCBI vs published QfO tree	conflict 0.000, nRF 0.128	conflict 0.000 (0/51 splits), nRF 0.1277	yes	`s1_conflict.py`, own bipartition code (ete3 only for parsing)

Not rerun (cost or scope): full OrthoBench/QfO pipeline runs (>30 min), S4 tree inference, S5 label building, S8 genome searches, QfO final scores (not in this task).

2. Discrepancies and concerns

1. **No numerical discrepancy found.** Every recomputed number matches to the reported precision.
2. **Undisclosed S0 split-v1 exposure (minor; disclosure gap).** Before the homology-aware v2 split, three S0 configs (`dc_a30_c50`, `deep_c20`, `deep_c50`, commit 6711002) were scored per-RefOG on the v1 "dev" half. 16 of those 35 RefOGs are now in v2 **test** (RefOG009, 014, 016, 017, 018, 022, 027, 031, 032, 036, 037, 047, 049, 056, 064, 069). Their per-RefOG completeness was in git. `docs/S0_families.md` §1 records the v1→v2 switch and the re-scoring, but `docs/DISCLOSURES.md` does not say that test RefOGs were seen. The impact is probably small: it affects only S0 stage-1 choices, and v2 was built by a fixed-seed mechanical rule. It should still be disclosed.
3. **Bootstrap reproduction is not an independent Monte Carlo draw.** My seed-11 CIs match the reported ones to about 1e-6 because both implementations use `default_rng(11).integers(0, n, (B, n))`. Other seeds and B=10,000 move the bounds by ≤0.7 pp, and no verdict changes.
4. **Commit-id inconsistency.** `results/freeze.json` gives `git_commit` 6c593a8 (the parent of the

tag). The tag `freeze-v1` is 71117bf, and test_log entries cite f7bcddc. `ancestra/` is identical in all of them and in HEAD, so the frozen code is unaffected, but freeze.json should name the tag commit. freeze.json hashes only `ancestra/*.py`. `ancestra/schemas/`, `ancestra_eval/` and the harness scripts (`results/orthobench_final/scripts/run_cached.py`) are not hashed (git shows no ancestra/ change).

5. **Runtime comparison is indicative only.** Ancestra ran with 10 threads and the competitors with 8. DESIGN §5 asks for 24 threads, and the node was shared. I did not check whether gene-order table building (GFF parsing) is included in Ancestra's 3.91 CPU-h (DESIGN §5: "all preprocessing counts"). "3.9 vs 8.2 CPU-h" should carry these caveats.
6. **Giant mixed root HOGs.** The OB accurate output has 6 root HOGs with ≥500 genes (5,840 genes in total; largest 1,910 at Amniota, likely olfactory receptors). Some have implausible copy profiles, e.g. HOG0010073 (Bilateria, 1,193 genes: opossum 532, dog 376, chicken 189, human 3). None contains a RefOG gene, so the OrthoBench score is unaffected, but pair-level precision in such families (QfO VGNC over-merging) is a known weakness. 20 random root HOGs (seed 11) look plausible: 11/20 have at most 2 copies per species, sizes 2–176, and one is an exact RefOG069 group (8/8). The rest are one larger multi-copy vertebrate family (176 genes, up to 31 copies) and in-paralog expansions (fly 17 copies, chimp 9).
7. **Wording (minor over-claim).** RESEARCH_TREE 8b2 (QfO peek 1) says "Precision comparable" and "ties FastOMA" on point estimates without a CI. Under DESIGN §5 this should read "inconclusive".
8. **S4 vs presets.** S4's pre-registered verdict was "keep NJ in localtree" (cost rule). The frozen `accurate` preset uses FastTree, adopted later on dev OG-F. That is legitimate, but REPORT should say the S4 cost verdict was superseded and why.
9. **Test-set reuse for ablations.** 11 configs were scored on OB test. All were logged before scoring, and the primary endpoint (accurate) was fixed at the freeze, so this is acceptable. The ablation CIs are unadjusted (no Holm correction, DESIGN §5 "Layers"). Report them as exploratory.

3. Files

`results/verification/`: `ob_rescore.py/.json` (task 1), `ob_bootstrap_sensitivity.json`, `official_cli_all70.txt`, `pytest_not_slow.log`, `toy_e2e.json`, `verify_orthoxml_accurate_go.{txt,json}`, `spotcheck_hogs.{py,txt,json}`, `spotcheck_20hogs.orthoxml`, `s0_completeness.{py,json}`, `s5_auc.{py,json}`, `s8_controls.json`, `s4_acc.{py,json}`, `s1_conflict.{py,json}`. Suggested CLAIMS status: the rows above marked "yes" can move to `verified` (verifier, 2026-10-27). That covers C-S0-2, C-S0-4, C-S1-1, C-S5-2 (human/mouse and the other pairs listed), C-S8-2 (recovery and negatives), C-CORE-1 (toy range) and the OrthoBench final rows. The PI decides.

Pass 2: final QfO results (results/qfo_final/, 2026-10-27 T+19h15 to T+19h22)

Same independence rules as pass 1. Code and outputs: `results/verification/qfo_*`.

4. Claim table (QfO)

claim	reported	recomputed	match?	how / notes
Provenance cand2_accurate / cand2_fast	freeze-v1 code	manifest `driver.code_sha256`: 24/24 == `freeze.json` for both; md5 of `hogs.orthoxml` == manifest outputs; the final QfO run read exactly these files (`work/qfo_final/<run>.log` input path)	yes	Manifest commits are d61da8b (accurate) and 7dec881 (fast). test_log says 394da46 / d61da8b. Same code, but the ids are inconsistent (see §5.5)
SwissTree TPR/PPV accurate	.721/.926	.7215/.9255 (±.037/.034)	yes	`qfo_refphylo.py`: own reimplementation of RefPhyloTest (truth parsed from `ReconciledTrees_*.drw`, own HOG flattening of `hogs.orthoxml`, protIds mapped with `mapping.json.gz`). Per-pair TP/FP/FN/TN counts are identical to the official raw file (3198/35/865/6318)
SwissTree fast	.731/.923	.7307/.9230	yes	counts identical to the official raw file
TreeFam-A accurate / fast	.583/.921, .608/.916	.5831/.9213, .6080/.9165	yes	counts identical (accurate 25,615 TP / 2,187 FP / 18,310 FN)
GO / EC accurate	.4748 (n 171,884) / .8985 (n 180,745)	.47484 / .89846, same n	yes	mean over the official raw per-pair files. 300/300 randomly sampled scored pairs per test are orthologs in my flattening of `hogs.orthoxml`
GO / EC fast	.4705 (181,857) / .8961 (196,729)	.47049 / .89606, same n	yes	300/300 pair-provenance check
Published competitor rows (20 tools incl. FastOMA, OMA HOGs / Pairs / Groups)	from OpenEBench	638/638 values in `summary.tsv` == `openebench_QfO_2022.tsv` (all benchmarks incl. STD/G-STD2 and n); no tool missing	yes	`qfo_competitors.py`. OMA rows are present in the main, STD and ablation tables

claim	reported	recomputed	match?	how / notes
Ablations used freeze-v1 code	yes	7/7 ablation manifests: 24/24 code hashes == freeze.json, git f7bcddc, not dirty; tree md5 in manifest == the arm's tree file; hogs md5 == manifest	yes	
Cache reuse only tree-independent stages	S0 + S2 only	`genes.tsv`, `families.npy`, `scores.npz`, `families_final.npy` byte-identical (the same inode) in det_nogo, cand2_* and all 7 ablations. merge/output are per-run (nlink 1). Driver `transplant()` re-keys only families/scores, checks that the key formula reproduces the source keys, and aborts on a code-key mismatch. The gene table is tree-independent (`load_proteomes` sorts by file name), and scoring takes no tree	yes	
Ablation c_astral SwissTree/TreeFam-A	.782/.929, .636/.914	.7821/.9288, .6360/.9138; counts identical to the official raw file	yes	
All QfO final evaluations logged before scoring	24 logged	g_accurate_nogo logged 09:48:13, its assess files written 09:53–09:54; FAS rerun logged as a rerun	yes	

5. Discrepancies and concerns (QfO)

1. **DISCLOSURES #1 is factually wrong.** It says test-only TreeFam-A families were "never scored separately before the freeze". `results/qfo_peek2/treefamA_dev_test.json` (commit 6c593a8, 06:05) contains TreeFam-A **test** subset values for both cand2 presets (accurate .580/.913, fast .605/.914). RESEARCH_TREE 8.0 quotes them. The freeze followed one minute later. The impact is small: the presets were declared before peek 2, and peek 2's pooled TreeFam-A already includes those families. The disclosure still needs correcting.

2. **The pre-registered primary endpoints are not the ones 8.2 reports.** DESIGN §5 names TreeFam-A **test**, SwissTree, **G-STD2-Eukaryota (RF with n)** and EC (with n). 8.2 reports pooled TreeFam-A (it includes the 30% dev families used for tuning) and STD-Eukaryota. It does not report G-STD2-Eukaryota at all. On G-STD2-Eukaryota Ancestra is worst among the main comparators: RF .257 (n 3,379) accurate and .266 (4,753) fast, against OrthoFinder .249 (6,629), FastOMA .239 (3,092), OMA HOGs .237 (2,981) and Hieranoid .222 (3,643). On EC accurate .899 (n 181k) is below OrthoFinder .931 (n 204k) on both axes.
3. **Non-inferiority under the pre-registered rule** (my paired bootstrap from the official raw files, B=2000, seed 11; over the 749 TreeFam-A test families, or over the 18 SwissTree cases; `qfo_primary_ni.json`). Accurate on TreeFam-A test: TPR .580 vs OrthoFinder .739, Δ −0.160 [−0.210, −0.106], so **it fails**. Vs FastOMA +0.043 [−0.005, +0.093] (NI on TPR). Vs OMA HOGs +0.002 [−0.058, +0.067] (inconclusive). Vs OMA Pairs +0.018 [−0.028, +0.069] (NI). PPV is not shown NI vs any comparator except OMA HOGs and OMA Pairs. SwissTree TPR vs OrthoFinder is −0.103 [−0.177, −0.029] (fails), and inconclusive vs the rest. 8.2 says "not on the Pareto front; TreeFam-A ≈ FastOMA/OMA; behind OrthoFinder on recall", which is consistent. But the formal verdict (fails vs OrthoFinder on both recall endpoints; NI vs FastOMA / OMA Pairs on TreeFam-A TPR only) should be stated.
4. **The ablation CIs are too narrow.** The official ± (TreeFam-A ±0.006) is a pair-level binomial CI that ignores the clustering of pairs within families. A paired bootstrap over the 1,108 families gives: ASTRAL-Pro tree +0.028 [−0.009, +0.077], FastME +0.022 [−0.001, +0.045], no gene order −0.012 [−0.028, +0.001], 5 NNIs +0.003 [−0.020, +0.026] (`qfo_ablation_familyboot.txt`). So RESEARCH_TREE 8.2's "gene order helps QfO recall a little" and any reading of the ASTRAL/FastME recall gains as significant are **not supported**. The random-tree control (−0.243) is far outside any CI.
5. **Over-statements in 8.2.** "STD comparable" has no CI. Accurate STD-Eukaryota .113±.006 is close to OrthoFinder (.118) but clearly worse than OMA HOGs .086, OMA Pairs .085, Hieranoid .074 and InParanoid .079. Under DESIGN §5 it should read "inconclusive vs OrthoFinder, worse than OMA / Hieranoid / InParanoid". "VGNC precision weak (vertebrate over-merging)": PPV .982 equals OMA Pairs (.980) and OMA HOGs (.979), and the over-merging cause is a hypothesis. The explanation for the EC drop with ASTRAL/FastME trees ("Bacteria+Archaea merged") is not tested here.
6. **Mislabelled error bars.** `results/qfo_final/summary.md` calls the GO/EC "±" values SEM. They are 1.96×SEM (GO accurate: raw SEM 0.000567, reported 0.00111).
7. **Commit ids** (as in §2.4). test_log peek-2 entries name 394da46 / d61da8b, but the manifests say d61da8b / 7dec881. The ablation entry for g_accurate_nogo names git=b310546, but its manifest says f7bcddc. Code hashes match freeze-v1 in every case, so this is a bookkeeping issue only.

Not rerun: the official `run_qfo_local.sh` (not needed, because the per-pair counts reproduce the official raw output exactly), STD/G-STD2 and FAS (not recomputed; competitor values checked against OpenEBench only).